Lab 3: Monte Carlo Sampling (2h)

CGR 2026

Instructions

- Please bring a ruler to this lab.
- Please ensure that you have downloaded this problem sheet during your lab slot.
- You are encouraged to read through the questions beforehand, but please leave the active solving for the session.
- Discussion is encouraged, but please write your own answers. Keep your answers concise.
- Submit handwritten answers to the tutor before the end of the lab. Show your working.
- Please **do not** nominate others to submit your work.
- Please put your name and student number on all sheets of your submission.
- You may use lecture material or web resources to look up information.
- You may **not use generative AI tools** to solve or check (before submission) answers.

Question 1: Estimating π Using a Monte Carlo Experiment

Working in pairs

In this experiment, you will estimate the value of π using a simple physical implementation of the Monte Carlo method. Each student will perform their own experiment and produce an independent estimate of π, while working with a partner to introduce randomness into the sampling process.

1. Each student will be given a printed A4 sheet containing a circle of a specified radius.
2. Roll a small piece of Blu Tack (will be provided) into a tiny ball. Hold it above your sheet, close your eyes, and drop it onto the paper. Your partner should gently jitter the sheet on the table to introduce additional randomness.
3. Mark the location where the Blu Tack lands using a pen or pencil. Label each sample with its trial number (e.g., write a small 1 next to the first sample, a 2 next to the second, and so on).
4. Repeat the process at least 50 times, recording the location and trial number of every sample on your sheet.
5. Swap roles with your partner. Repeat the experiment using your partner's printed circle, with you now responsible for jittering the sheet.

After completing the experiment, answer the following questions. Record your answers on the back of your printed sheet. **Note:** Although the experiment is conducted in pairs, each student must maintain their own set of measurements, calculations, plot, and conclusions.

(a) Using your recorded sample locations, estimate π using the Monte Carlo method described in class. Show how you obtained your estimate.

(b) Plot your estimated value of π against the trial number. Your plot should show how the estimate changes as the number of samples increases. An approximate plot, using a standard ruler for measurement is sufficient.

(c) State your final estimate of π. Why does your estimate differ from the true value? (at most two sentences)

(d) Identify possible sources of experimental error in this procedure and their expected impacts (a concise bulleted list will suffice).

(e) Would it converge to the correct value if you used infinite samples? Explain the reason in one sentence.

(f) Compare your results with those of your partner. List any differences in the error or plots.

(g) Write down the radii of your circle and your partner's. Which experiment would you expect to produce a higher estimation error? Explain the reason inone sentence)?

Question 2: Monte Carlo Estimation of Irradiance

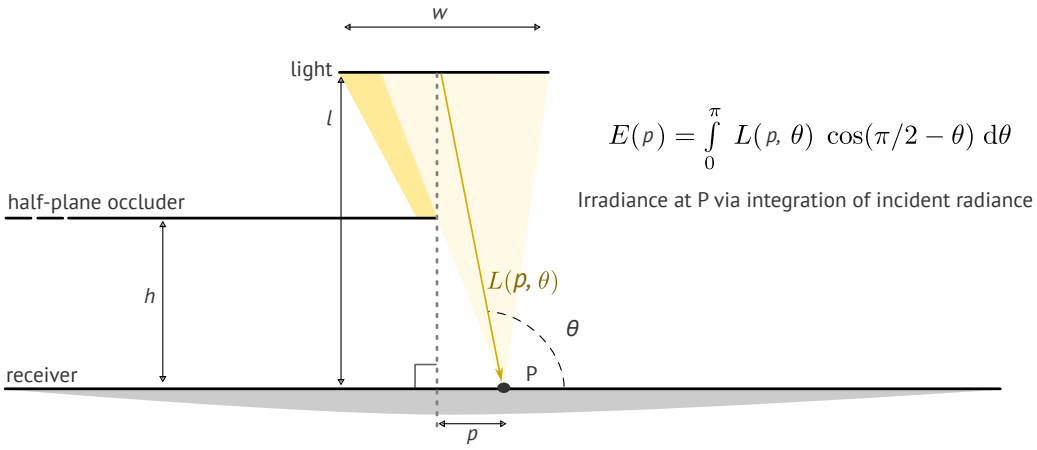


Figure 1: 2D scene for irradiance estimation.

Consider the 2D scene shown in Figure 1. It consists of a line emitter with unit radiance, an occluding half-plane and a receiving plane along the x-axis. The line emitter is centred at $x = 0$, has width w, and is located at height l above the receiving plane. The occluding half-plane is located at height h and extends towards negative x, ending at $x = 0$. Your task is to estimate the irradiance at a point $P = (p, 0)$, where $p > 0$. The irradiance can be calculated by integrating over the range of directions for which the emitter is visible from P. To achieve this, take the following steps.

1. Using the geometry of the scene, determine the angular interval over which the emitter is visible. Express your answer in terms of w, p, h and l.
2. Tabulate the x-coordinates of the Blu Tack samples recorded in Question 1. Map these values to the the domain of integration $[0, \pi]$. Write down the true integral sought and the Monte Carlo estimator for irradiance at P via uniform angular sampling over the interval $[0, \pi]$.
3. What contribution does each random sample makes to the estimate?
4. How do you account for rays occluded by the half-plane?
5. How are the individual samples contributions are combined to obtain the final estimate. Then use your random numbers to calculate the Monte Carlo estimate of the irradiance at P. Show your working.

The above procedure uses the generated samples over $[0, \pi]$ to estimate the irradiance at P. This could be inefficient because many of the samples are occluded. Rewrite the integral in terms of the angular interval over which the emitter is visible θ_1, θ_2. Then, write down the Monte Carlo estimator for the irradiance at P using samples mapped to $[\theta_1, \theta_2]$ instead of $[0, \pi]$. Use the same random numbers from Q1 in this estimator for the irradiance at P by. Compare this with the previous estimate. Which one do you expect to be more accurate? Explain your reasoning in one sentence.